

Improved Prediction of Drug–Target Interactions Using Self-Paced Learning with Collaborative Matrix Factorization

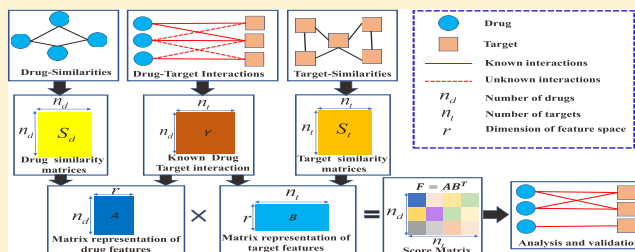
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Supporting Information

ABSTRACT: Identifying drug–target interactions (DTIs) plays an important role in the field of drug discovery, drug side-effects, and drug repositioning. However, in vivo or biochemical experimental methods for identifying new DTIs are extremely expensive and time-consuming. Recently, in silico or various computational methods have been developed for DTI prediction, such as ligand-based approaches and docking approaches, but these traditional computational methods have several limitations. This work utilizes the chemogenomic-based approaches for efficiently identifying potential DTI candidates, namely, self-paced learning with collaborative matrix factorization based on weighted low-rank approximation (SPLCMF) for DTI prediction, which integrates multiple networks related to drugs and targets into regularized least-squares and focuses on learning a low-dimensional vector representation of features. The SPLCMF framework can select samples from easy to complex into training by using soft weighting, which is inclined to more faithfully reflect the latent importance of samples in training. Experimental results on synthetic data and five benchmark data sets show that our proposed SPLCMF outperforms other existing state-of-the-art approaches. These results indicate that our proposed SPLCMF can provide a useful tool to predict unknown DTIs, which may provide new insights into drug discovery, drug side-effect prediction, and repositioning existing drug.



INTRODUCTION

Computational prediction of drug–target interactions (DTIs) plays an important role in the field of drug discovery,¹ drug side-effect prediction,² and drug repositioning,³ aiming to discover novel targets or novel drugs interacting with existing drugs.^{4–6} However, in vivo or biochemical experimental methods for identifying new DTIs are extremely expensive and time-consuming. To improve the drug discovery efficiency, various computational methods have been developed to efficiently identify potential DTI candidates, which can significantly reduce the time and cost. According to previous works,⁴ they can be divided into three groups: ligand-based approaches, docking approaches, and chemogenomic approaches. The first group is ligand-based approaches, which predicts the new DTI by using similarity between the proteins' ligands because similar drugs share similar properties and bind similar target proteins.^{7,8} However, this approach may not be effective with the small number of known ligands per protein.⁹ The second group is docking approaches, which are widely used in biology. This kind of docking approach involves interactions of drugs and targets by utilizing the 3D structures of drugs and proteins. Nevertheless, the docking approaches have two problems: (1) it is still challenging to take 3D structures of many targets,^{10,11} especially G protein-coupled receptors (GPCR), and (2) the process of simulation is time-consuming. The third group is chemo-

genomic methods, which are effective and efficient ways for DTI prediction and allow larger-scale predictions within a very short period of time, so as to provide candidate drugs or target proteins for experimental screening.

In recent years, a lot of effort has been made to develop chemogenomic methods for DTI prediction. They can be classified into three types: classification-based methods, network diffusion-based (network propagation) methods, and matrix factorization (MF)-based methods.^{4,12–14}

Recently, classification-based methods were used to predict unknown DTIs. For example, Bleakley et al.¹⁵ developed a supervised approach for predicting DTIs using a support vector machine based on a bipartite local model. Perlman et al.¹⁶ utilized similarity-based inference of drug–targets (SITAR), which consists of a new scoring scheme for drug–target associations based on a given pair of drug–drug and target–target similarity measures and performed DTI prediction using a logistic regression component. Shi et al.¹⁷ enhanced the similarity measures to include nonstructural information and introduced the concept of a “supertarget” to handle the problem of possibly missing interactions. Shi et al.¹⁸ characterized each drug–target pair as a feature vector of within-scores and

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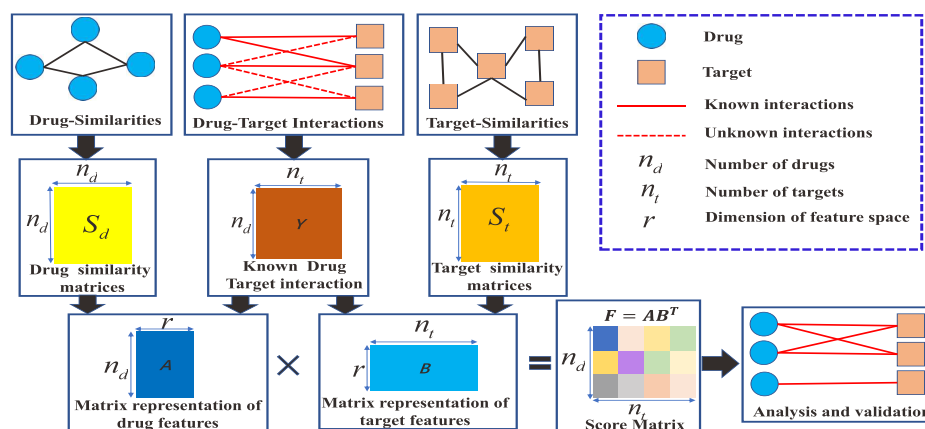


Figure 1. Diagram illustrating the process of our proposed SPLCMF.

between-scores. Chen et al.¹⁹ proposed NetCBP for DTI prediction based on the assumption that similar drugs often target similar proteins. Shi et al.²⁰ proposed the Spy and Supertarget, introduced accordingly and further integrated to form a two-layer local classification model, for missing interactions. Then, in order to improve the DTI prediction accuracy, network diffusion-based (network propagation) techniques were incorporated to predict unknown DTIs. For example, Cheng et al.²¹ used drug-based similarity inference, target-based similarity inference, and network-based inference (NBI) based on complex network theory to predict DTIs. Seal et al.²² applied the random walk with restart (RWR) method to a heterogeneous network of drugs and targets compiled from the DrugBank database. Wu et al.¹⁴ used network-based methods, which do not rely on three-dimensional structures of targets and a negative sample. Zhang et al.²³ proposed a label propagation method with linear neighborhood information (LPLNI) to predict unobserved DTIs. Xia et al.²⁴ defined a novel approach, namely, Laplacian regularized least-squares (LapRLS), which uses regularized least-squares to minimize the objective function including an error term and graph regularization term as the local model.

Finally, most existing prediction methods mainly focus on exploiting information from MF⁵ methods, where MF factorizes a matrix into two low-rank matrices, which correspond to feature spaces of drugs and targets. For example, Gönen et al.²⁵ proposed kernelized Bayesian matrix factorization with twin kernels (KBMF2K), which uses dimensionality reduction, MF, and binary classification to predict DTIs. Zheng et al.²⁶ developed multiple similarities collaborative matrix factorization (MSCMF), which integrates multiple kinds of drug similarities and target similarities to CMF for DTI prediction. Liu et al.²⁷ used neighborhood regularized logistic matrix factorization (NRLMF), which focuses on modeling the probability that a drug would interact with a target by logistic MF. Cobanoglu et al.²⁸ introduced probabilistic matrix factorization (PMF), which predicts DTI interactions using probabilistic linear models with Gaussian noise. Ezzat et al.²⁹ utilized graph regularized matrix factorization (GRMF) and weighted graph regularized matrix factorization (WGRMF), which add graph regularization terms to learn a manifold for label propagation. Shi et al.³⁰ utilized a triple matrix factorization (TMF) to capture the relationship between drug–target pairs in the latent pharmacological space, where TMF has capacity to build a unified solution of predicting DTI. However, the main limitation of existing MF methods is the nonconvexity of their objective functions. It is easy to make

current MF methods fall into bad local minima, especially in the presence of heavy noises and missing data. The most commonly used methods are to implement algorithms with different initializations and choose an optimal solution among them. However, it is not suitable for an unsupervised setting because there is no straightforward criterion for choosing a proper solution.³¹

To address the above challenges of MF, we propose a novel MF framework, namely self-paced learning with collaborative matrix factorization based on weighted low rank approximation (SPLCMF). CMF projects drugs and targets into a common low-rank feature space and then integrates multiple networks related to drugs and targets into regularized least-squares. Moreover, some previous research^{32–34} has demonstrated that SPL has capacity to avoid bad local minima and achieve better generalization and is practically effective in the presence of heavy noises and missing data. SPL,³² inspired by the learning process of humans or animals, gradually incorporates the training samples into learning in small or large samples. Instead of the curriculum learning approach, where the data is located in a predefined order, SPL learns the training data in an order from simple to complex, dynamically determined by the feedback of the learner. Thus, we integrate SPL into CMF to improve DTI prediction performance. Furthermore, the SPLCMF framework can select samples into training by using soft weighting, which tends to reflect the potential importance of the samples in training because the noises embedded in data are generally nonhomogeneous across the sample. The flow diagram in Figure 1 shows the process of our proposed SPLCMF framework to improve the DTI prediction performance.

Then, we applied ChEMBL, DrugBank, and KEGG to validate the new DTIs, where some drug–target pairs are reported in at least one of these databases. These results indicate that SPLCMF can provide a useful tool to predict new DTIs and repurpose an existing drug so as to increase research productivity toward drug repositioning and polypharmacology.

MATERIALS

In this study, five benchmark data sets are utilized to validate our proposed SPLCMF method for DTI predictions, including nuclear receptors (NR), GPCR, ion channels (IC), and enzymes (E). These data sets were originally provided by Yamanishi et al.³⁵ Another data set used in this work was obtained from the work of Kuang et al.³⁶ Brief descriptions of drugs, targets, and DTIs from different data sets are given in Table 1.

Table 1. Brief Descriptions for the DTI Networks

data sets	Yamanishi ³⁵				Kuang ³⁶
	NR	GPCR	IC	E	—
number of drugs	54	223	210	445	786
number of targets	26	95	204	664	809
number of DTIs	90	635	1476	2926	3681
average degree of drugs	1.67	2.85	7.03	6.58	4.68
average degree of targets	3.46	6.68	7.24	4.41	4.55
sparsity of the interaction matrix	93.59%	97.00%	96.55%	99.01%	99.42%

METHODS

In this section, we introduced a novel MF framework, namely, self-paced learning with CMF based on weighted low-rank approximation (SPLCMF), to identify unknown DTIs pairs.

Notation. Suppose we have a set of known drugs $\mathbf{D} = \{d_1, d_2, \dots, d_{N_d}\}$ and known targets $\mathbf{T} = \{t_1, t_2, \dots, t_{N_t}\}$, where N_d is the number of drugs and N_t is the number of targets, respectively. Let $\{S_d, S_t\}$ be a drug similarity matrix and target similarity matrix. The dimensions of S_d and S_t are $N_d \times N_d$ and $N_t \times N_t$, respectively. Let $\mathbf{Y}$ be an $N_d \times N_t$ adjacency matrix, which is a known interaction between drugs and targets. If a drug d_i interacts with a target protein t_j , $y_{ij} = 1$, and $y_{ij} = 0$ otherwise. The purpose of our work is to evaluate the $\mathbf{F}$, which is an $N_d \times N_t$ score matrix. The score F_{ij} of $\mathbf{F}$ indicates that drug d_i interacts with target protein t_j .

Model Formulation. The purpose of MF is to factorize $\mathbf{Y}$ into two low-rank matrices $\mathbf{A}$ and $\mathbf{B}$, which correspond to feature spaces of drugs and targets, respectively. Then, we use an inner product between $\mathbf{A}$ and $\mathbf{B}$ to approximate DTI $\mathbf{Y}$ as follows

$$\mathbf{Y} \approx \mathbf{A}\mathbf{B}^T \quad (1)$$

where $\mathbf{A}$ and $\mathbf{B}$ are $N_d \times r$ and $N_t \times r$ and r is the dimension of the feature space.

The $\mathbf{A}$ and $\mathbf{B}$ could be estimated so that we could reconstruct $\mathbf{Y}$. That is to say, the unknown interaction pair could be assigned a score. The process of eq 1 is shown in Figure 2. In order to obtain $\mathbf{A}$ and $\mathbf{B}$, we apply minimization to the squared error in eq 1 as follows

$$\arg \min_{\mathbf{A}, \mathbf{B}} \|\mathbf{Y} - \mathbf{A}\mathbf{B}^T\|_F^2 \quad (2)$$

where $\|\cdot\|_F^2$ is the Frobenius norm.

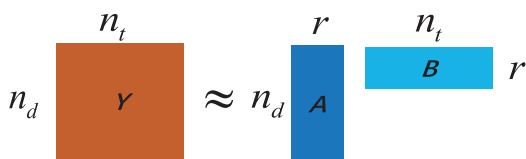


Figure 2. Process of MF.

To distinguish between known drug–target pairs (as training samples) and unknown drug–target pairs (as testing samples), we introduce an $N_d \times N_t$ weight matrix $\mathbf{W}$, which takes a value of 0 or 1; $W_{ij} = 1$ indicates that Y_{ij} is known pairs (an interacting or noninteracting pair); otherwise, $W_{ij} = 0$. Therefore, eq 2 can be rewritten as

$$\arg \min_{\mathbf{A}, \mathbf{B}} \|\mathbf{W} \odot (\mathbf{Y} - \mathbf{A}\mathbf{B}^T)\|_F^2 \quad (3)$$

where $\mathbf{W} \odot \mathbf{Z}$ denotes the element-wise product of matrices $\mathbf{W}$ and $\mathbf{Z}$.

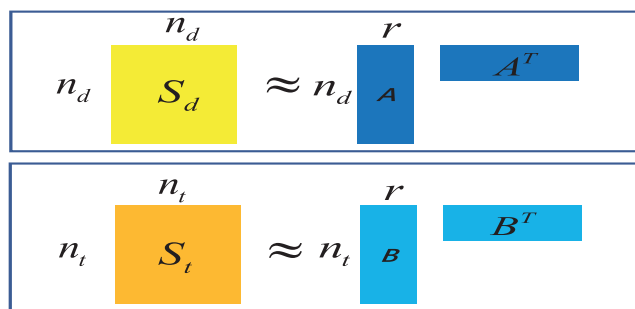
Directly solving eq 2 can result in overfitting in training data. Therefore, to solve the above issue, we add the L_2 regularization terms to eq 3

$$\arg \min_{\mathbf{A}, \mathbf{B}} \|\mathbf{W} \odot (\mathbf{Y} - \mathbf{A}\mathbf{B}^T)\|_F^2 + \lambda_1(\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2) \quad (4)$$

where λ_1 is a regularization parameter.

In order to integrate the drug similarity network, target similarity network, and known DTI networks into regularized least-squares, we use CMF. The idea of CMF is that DTIs $\mathbf{Y}$ should be close to the inner product between $\mathbf{A}$ and $\mathbf{B}$, the as same as MF. Inspired by this idea, the drug similarity matrix S_d and target similarity matrix S_t should be approximated by the inner product of the corresponding two drug feature vectors and two target feature vectors, respectively. The process of eq 5 is shown in Figure 3.

$$S_d \approx \mathbf{A}\mathbf{A}^T \quad S_t \approx \mathbf{B}\mathbf{B}^T \quad (5)$$

Figure 3. Process of MF for similarity matrix S_d and target similarity matrix S_t .

Thus, we apply similarity matrix S_d and target similarity matrix S_t to eq 4. The new formula is given as follows

$$\arg \min_{\mathbf{A}, \mathbf{B}} \|\mathbf{W} \odot (\mathbf{Y} - \mathbf{A}\mathbf{B}^T)\|_F^2 + \lambda_1(\|\mathbf{A}\|_F^2 + \|\mathbf{B}\|_F^2) + \lambda_d \|\mathbf{S}_d - \mathbf{A}\mathbf{A}^T\|_F^2 + \lambda_t \|\mathbf{S}_t - \mathbf{B}\mathbf{B}^T\|_F^2 \quad (6)$$

where λ_d and λ_t are regularization parameters.

Due to the presence of noise and missing data, most of the existing MF models are generally nonconvex, which can easily fall into bad local minima. To address the above issues, we integrate self-paced learning into CMF to improve the DTI prediction. Therefore, eq 6 can be rewritten as

$$\arg \min_{A,B,W} \|W \odot (Y - AB^T)\|_F^2 + \lambda_1(\|A\|_F^2 + \|B\|_F^2) + \lambda_d \|S_d - AA^T\|_F^2 + \lambda_t \|S_t - BB^T\|_F^2 - \frac{1}{k} W \quad (7)$$

According to Kumar et al.,³² with fixed **A** and **B**, the optimal W_{ij}^* is easily calculated by

$$W_{ij}^* = \begin{cases} 1 & \text{if } l_{ij} \leq 1/k \\ 0 & \text{otherwise} \end{cases} \quad (8)$$

Here, we explain eq 8, namely, hard weighting.³⁷ In the SPL iteration, when l_{ij} is less than the model “age” $1/k$, a sample, which is taken as a simple sample with smaller losses, can be selected ($W^* = 1$) in training. Otherwise, it is unselected ($W^* = 0$). As the model age of $1/k$ increases, it means that more likely hard samples with larger losses will be considered to train a more “mature” model. However, hard weighting does not incline to reflect the latent importance of samples in training because the noises embedded in data are generally inhomogeneous across samples. Instead of hard weighting, we apply soft weighting to eq 7

$$\arg \min_{A,B,W} \|W \odot (Y - AB^T)\|_F^2 + \lambda_1(\|A\|_F^2 + \|B\|_F^2) + \lambda_d \|S_d - AA^T\|_F^2 + \lambda_t \|S_t - BB^T\|_F^2 + \frac{\gamma^2}{W + \gamma k} \quad (9)$$

where k is the model age and γ is a parameter for the strength of the weight assigned to the selected sample.

Inspired by Zhao et al.,³¹ with fixed **A** and **B**, the optimal W_{ij}^* is easily calculated by

$$W_{ij}^* = \begin{cases} 1 & \text{if } l_{ij} \leq \frac{1}{(k + 1/\gamma)^2} \\ 0 & \text{if } l_{ij} \geq \frac{1}{k^2} \\ \gamma \left(\frac{1}{\sqrt{l_{ij}}} - k \right) & \text{otherwise} \end{cases} \quad (10)$$

Here, for E.q.(10), namely soft weighting, when l_{ij} is less than $\frac{1}{\sqrt{k}}$, the sample is treated as an easy sample and assigned nonzero weight. Besides, when l_{ij} is less than $\frac{1}{\sqrt{k+1/\gamma}}$, the corresponding sample is treated as a faithfully easy sample weighted by 1. Finally, in other situations, the sample is treated as a hard sample weighted by 0. Figure 4 shows a simple example for hard weighting and soft weighting. The effectiveness of the proposed self-paced regularizer is evaluated in the Experimental Settings section.

Alternative Search Strategy for SPLCMF. Inspired by Kumar et al.,³² we utilize alternative search strategy (ASS)³⁸ to solve the objective function of SPLCMF. ASS is an iterative method that solves the optimization problem by dividing variables into two disjoint blocks and alternately optimizing each block with another fixed block. In the SPLCMF process, under fixed **A** and **B**, **W** can be calculated by

$$\arg \min_W \|W \odot (Y - AB^T)\|_F^2 + \frac{\gamma^2}{W + \gamma k} \quad (11)$$

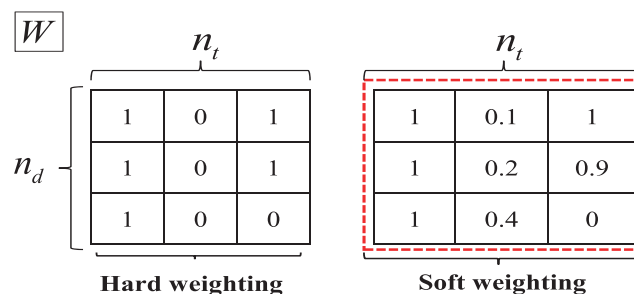


Figure 4. Comparison of hard weighting and soft weighting on the interacting or noninteracting pair. All pairs are used as positive in reranking. Soft weighting is learned by our proposed method.

where $l = \|(Y - AB^T)\|_F^2$ calculated under the current **A** and **B**. This optimization can be easily solved by eq 10. When **W** is fixed, **A** and **B** are calculated by eq 6, which utilizes the alternating least-squares (ALS) algorithm³⁹ to estimate **A** and **B**. In order to implement the ALS algorithm, we have rewritten eq 6 as follows

$$L = \sum_{i=1}^{N_d} \sum_{j=1}^{N_t} W_{ij} (Y_{ij} - a_i b_j^T)^2 + \lambda_1 \left(\sum_{i=1}^{N_d} \|a_i\|^2 + \sum_{j=1}^{N_t} \|b_j\|^2 \right) + \lambda_d \sum_{i=1}^{N_d} \sum_{p=1}^{N_d} (S_d(d_i, d_p) - a_i a_p^T)^2 + \lambda_t \sum_{j=1}^{N_t} \sum_{q=1}^{N_t} (S_t(t_j, t_q) - b_j b_q^T)^2 \quad (12)$$

where a_i and b_j are the i th and j th row vectors of **A** and **B**, respectively.

To minimize the a_i and b_j of L , we could take the partial derivative of L for a_i and b_j . The updating rules of **A** and **B** are obtained by setting $\frac{\partial L}{\partial a_i} = 0$ and $\frac{\partial L}{\partial b_j} = 0$.

The updating rules of **A** and **B** are given as follows

$$a_i = \frac{\sum_{j=1}^{N_t} W_{ij} Y_{ij} b_j + \lambda_d \sum_{p=1}^{N_d} S_d(d_i, d_p) b_p}{\sum_{j=1}^{N_t} W_{ij} b_j^T b_j + \lambda_1 I_k + \lambda_d \sum_{p=1}^{N_d} a_p^T a_p} \quad (13)$$

$$b_j = \frac{\sum_{i=1}^{N_d} W_{ij} Y_{ij} a_i + \lambda_t \sum_{q=1}^{N_t} S_t(t_j, t_q) a_q}{\sum_{i=1}^{N_d} W_{ij} a_i^T a_i + \lambda_1 I_k + \lambda_t \sum_{q=1}^{N_t} b_q^T b_q} \quad (14)$$

where I_k is the identity matrix.

Algorithm 1 illustrates the process of ALS for estimating each parameters. During implementation of the algorithm, we first initialize **A** and **B** randomly. Then, **A** and **B** are updated by eqs 13 and 14 until convergence. Finally, the score matrix **F** is calculated by

$$\mathbf{F} \doteq AB^T \quad (15)$$

F_{ij} is the predicted score between d_i and t_j . The higher the score, the higher the possibility of DTI. We analyze the time complexity of Algorithm 1. Suppose $iter1$ denotes the number of iterations, m is the number of drugs, n is the number of targets, and r is the feature space. To update **A** using eq 13, we need the time of $r^2 \cdot ((m+n)^2 + rm)$. To update **B** using eq 14, we need the time of $r^2 \cdot ((m+n)^2 + rn)$. Thus, the time complexity of Algorithm 1 is $O(iter1 \cdot (r^2 \cdot (m+n)^2 + r^3 \cdot (m+n)))$.

Therefore, The whole process of SPLCMF is summarized in Algorithm 2. The time complexity of Algorithm 2 is $O(iter2 \cdot iter1 \cdot (r^2 \cdot (m+n)^2 + r^3 \cdot (m+n)))$, where $iter2$ denotes the number of iterations for the outer loop, aiming to update **W**.

Algorithm 1 Pseudocode of parameter estimation for CMF

Require: Y : true drug-target interaction matrix; W : weight matrix; S_d and S_t : drug and target similarity matrices; r : the feature space; λ_t , λ_d and λ_i : regularization parameters
Ensure: A , B and score matrix F
 1: initial A and B randomly;
 2: **repeat**
 3: Update A using E.q.(13);
 4: Update B using E.q.(14);
 5: Update F using E.q.(15);
 6: **until** convergence

Algorithm 2 Pseudocode of parameter estimation for SPLCMF

Require: Y : true drug-target interaction matrix; S_d and S_t : drug and target similarity matrices; r : the feature space; λ_t , λ_d and λ_i : regularization parameters; $\mu > 1$: step size; k_0, k_{end}
Ensure: score matrix F
 1: Initial: solve the CMF problem with all the observation equally weighted to obtain A_0 and B_0 , calculate $\ell_{ij}, t \leftarrow 0, k \leftarrow k_0$
 2: **while** $k > k_{end}$ **do**
 3: Update W using E.q.(10);
 4: Update A and B using Algorithm 1;
 5: Update F using E.q.(15)
 6: Compute current $\ell_{ij}(Y - F)$
 7: $t \leftarrow t + 1, k \leftarrow k/\mu$
 8: **end while**

RESULTS

In order to evaluate the effectiveness of SPLCMF, we present the results of synthetic data and five benchmark data sets.

Several Metrics. To assess the performance of our proposed SPLCMF method, several metrics are adopted, i.e., root-mean-square error (RMSE), mean absolute error (MAE), the area under the ROC curve (AUC), and the area under the precision-recall curve (AUCPR). Of note, there is a problem of class imbalance in the benchmark data sets, where AUCPR is suitable to evaluate the performance for each method.⁴⁰

Synthetic Data. In this section, the competing methods are CMF and SVD, designed to handle missing data. Inspired by Zhao et al.³¹ and Zheng et al.,⁴¹ the matrix Y' was generated randomly with $m = 200$, $n = 300$, and $r = 3$, which is used by Gaussian distribution $N(0, 1)$. Then we consider the cases with the different missing ratios (10, 50, 90%) and noise ratios (5, 10, 20, 25, 40%). The noise ratios of the entries of Y' were added to uniform noise on $[-20, 20]$. According to the work,⁴² the Y' can be transformed into Y because the elements of the matrix Y' and the matrix Y are continuous values and binary values, respectively. Finally, the experiments are implemented 30

times for each case. Besides, we adopt two criteria, such as, $RMSE = \frac{1}{\sqrt{mn}} \|Y' - AB^T\|_F$ and $MAE = \frac{1}{mn} \|Y' - AB^T\|_1$, to assess the performance of two methods, where m and n are the rows and columns of the matrix Y . The performance of each method was evaluated in terms of these two criteria as the average over the 30 realizations, exhibited in Table 2.

As shown in Table 2, the results are obtained by CMF, SVD, and SPLCMF with different missing levels and noise levels. For each case, our proposed SPLCMF method achieves the best performance among all methods in terms of RMSE and MAE. For example, when missing_ratio = 10% and noise_ratio = 5%, the RMSE and MAE of SPLCMF are 0.746 and 0.222, much better than 1.34 and 0.497 and 1.804, and 0.755 obtained by CMF and SVD. When the missing ratio decreases, the prediction performances of the two methods are improved. For example, when noise_ratio = 40%, the MAE of the SPLCMF is 2.94 and 1.462 with the missing_ratio = 90 and 50%. When the noise ratio increases, the prediction performances are decreased among methods. For example, when missing_ratio = 90%, the RMSE of SPLCMF is from 0.996 to 3.54, in which the noise ratio increases from 5 to 40%. In a word, our SPLCMF method outperforms CMF and SVD in term of noise and missing data in simulation data.

Benchmark Data Sets. In this section, we analyze the influences of parameters on the SPLCMF model, which has four parameters: the dimension of feature space r and the regulation parameters λ_t , λ_d , and λ_i . Besides, several methods are compared with our proposed SPLCMF method: SITAR,¹⁶ nearest profile (NP), weighted profile (WP),³⁵ NBI,²¹ LapRLS,²⁴ KBMF2K,²⁵ and CMF.²⁶ Finally, several metrics are adopted to evaluate the prediction performances, including AUC and AUCPR.

Experiments Settings. Following previous works,^{4,12,24–27,42–44} we use 10-fold cross-validation 5 times to assess the performance of DTI prediction methods. The data set is divided into two groups with random sampling (90% training data set and the remaining 10% test data set). This process is repeated 10 times. Then, in order to avoid errors caused by random partition of the data set, the DTI prediction methods are implemented under 5 trials of 10-fold cross-validation. Finally,

Table 2. Performance Comparison of CMF, SVD, and SPLCMF in Terms of RMSE and MAE on Synthetic Data^a

missing_ratio (%)	noise_ratio (%)	MAE			RMSE		
		CMF	SVD	SPLCMF	CMF	SVD	SPLCMF
10	5	0.497(0.040)	0.755(0.048)	0.222(0.005)	1.340(0.040)	1.804(0.048)	0.746(0.029)
	10	0.667(0.026)	0.935(0.034)	0.296(0.009)	1.472(0.038)	1.970(0.034)	0.888(0.031)
	20	0.864(0.023)	1.159(0.048)	0.432(0.016)	1.635(0.038)	2.164(0.048)	1.078(0.042)
	25	0.930(0.022)	1.246(0.045)	0.514(0.020)	1.694(0.026)	2.230(0.045)	1.189(0.043)
	40	1.113(0.025)	1.481(0.049)	0.871(0.032)	1.833(0.036)	2.411(0.049)	1.712(0.060)
50	5	0.681(0.039)	0.795(0.045)	0.259(0.008)	1.776(0.047)	2.153(0.045)	0.889(0.047)
	10	0.911(0.033)	1.018(0.038)	0.351(0.013)	1.970(0.052)	2.346(0.038)	1.075(0.053)
	20	1.151(0.022)	1.297(0.031)	0.552(0.022)	2.231(0.037)	2.565(0.031)	1.393(0.063)
	25	1.228(0.032)	1.411(0.038)	0.659(0.032)	2.289(0.051)	2.650(0.038)	1.554(0.085)
	40	1.462(0.026)	1.734(0.042)	1.094(0.037)	2.469(0.042)	2.889(0.042)	2.157(0.078)
90	5	0.656(0.027)	0.775(0.078)	0.402(0.019)	2.453(0.072)	2.571(0.078)	0.996(0.059)
	10	1.247(0.035)	1.138(0.034)	0.497(0.017)	3.315(0.080)	2.881(0.034)	1.337(0.072)
	20	2.027(0.037)	1.514(0.032)	0.890(0.042)	4.262(0.064)	3.186(0.032)	2.171(0.120)
	25	2.307(0.036)	1.683(0.037)	1.110(0.045)	4.559(0.083)	3.322(0.037)	2.520(0.128)
	40	2.940(0.052)	2.138(0.044)	1.846(0.079)	5.129(0.085)	3.649(0.044)	3.540(0.148)

^aThe results are averaged over 30 runs, and the best results are highlighted in bold. SDs are given in parentheses.

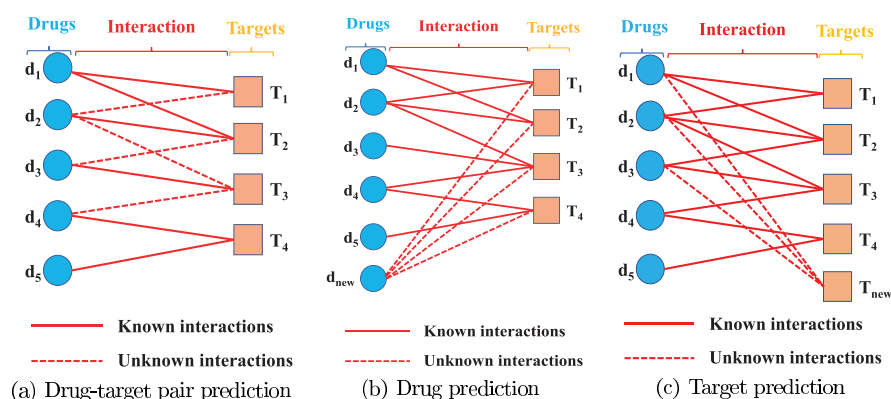


Figure 5. Three different cross-validation settings conducted for each method.

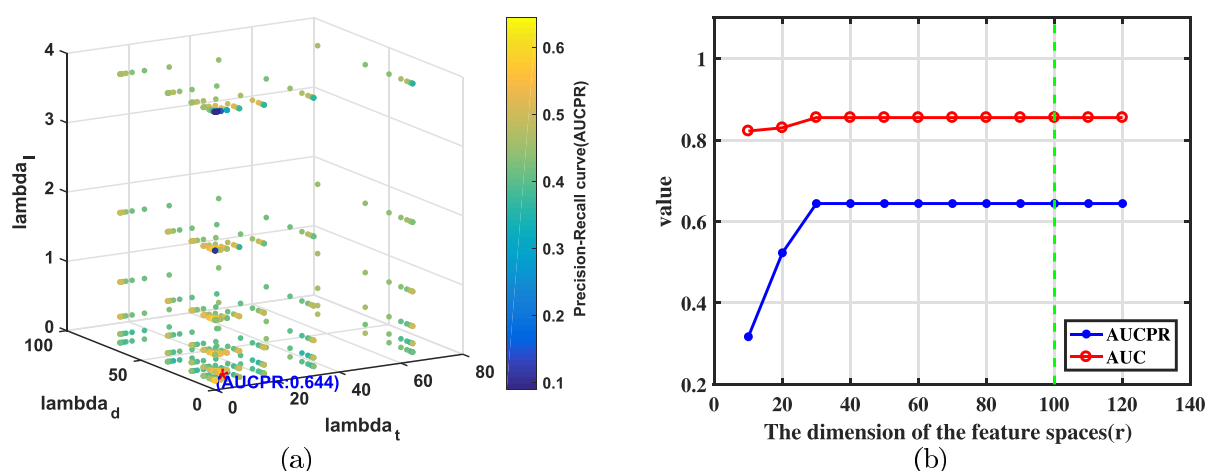


Figure 6. Influences of parameters on the SPLCMF model on the NR data set under CVS1. (a) AUCPRs are obtained by different combinations of λ_d , λ_t , and λ_1 . The optimal hyperparameters are $\lambda_1 = 0.125$, $\lambda_d = 0.125$, and $\lambda_t = 0.125$. (b) AUCs and AUCPRs are obtained by assigning different r values.

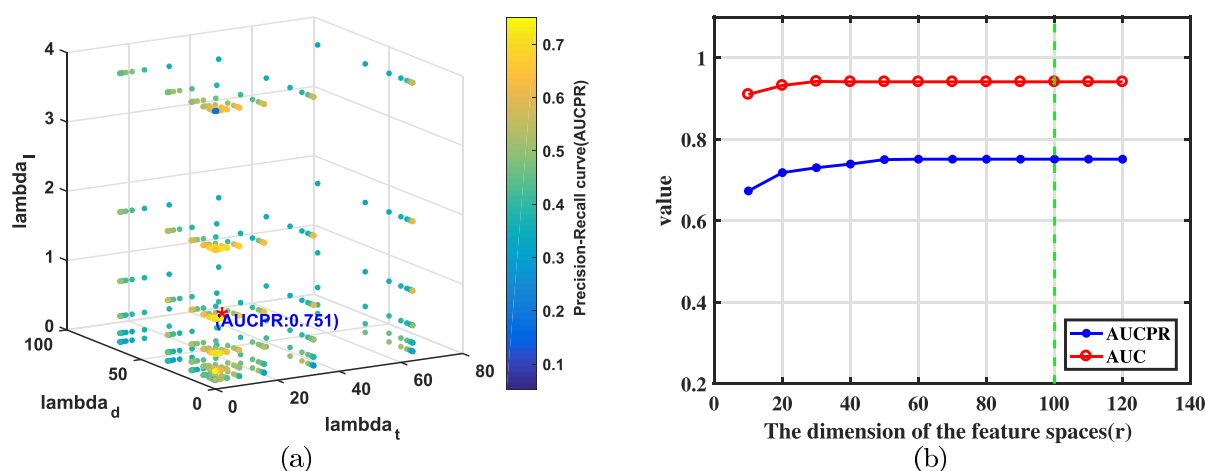


Figure 7. Influences of parameters on the SPLCMF model on the GPCR data set under CVS1. (a) AUCPRs are obtained by different combinations of λ_d , λ_t , and λ_1 . The optimal hyperparameters are $\lambda_1 = 1$, $\lambda_d = 0.5$, and $\lambda_t = 0.25$. (b) AUCs and AUCPRs are obtained by assigning different r values.

the AUCs and AUCPRs of the 10-fold CVs are averaged to give the final AUC and AUCPR for each method.

Inspired by previous study,^{4,45} three different cross-validation settings are analyzed for each method.

CVS1: In Figure 5a, a known drug may be interacted with one known target. During experiments, in each round, 90% of

elements in Y are regarded as training data and the remaining 10% of the elements as test data.

CVS2: In Figure 5b, a new drug may be interacted with one or several targets. During experiments, in each round, 90% of rows in Y are regarded as training data and the remaining 10% of rows as test data.

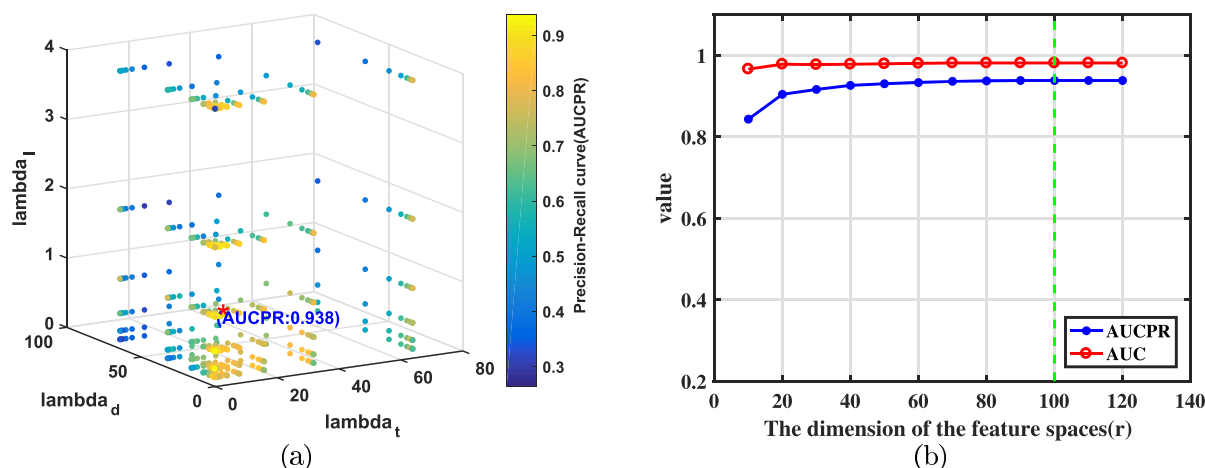


Figure 8. Influences of parameters on the SPLCMF model on the IC data set under CVS1. (a) AUCPRs are obtained by different combinations of λ_l , λ_d , and λ_t . The optimal hyperparameters are $\lambda_l = 1$, $\lambda_d = 1$, and $\lambda_t = 0.5$. (b) AUCs and AUCPRs are obtained by assigning different r values.

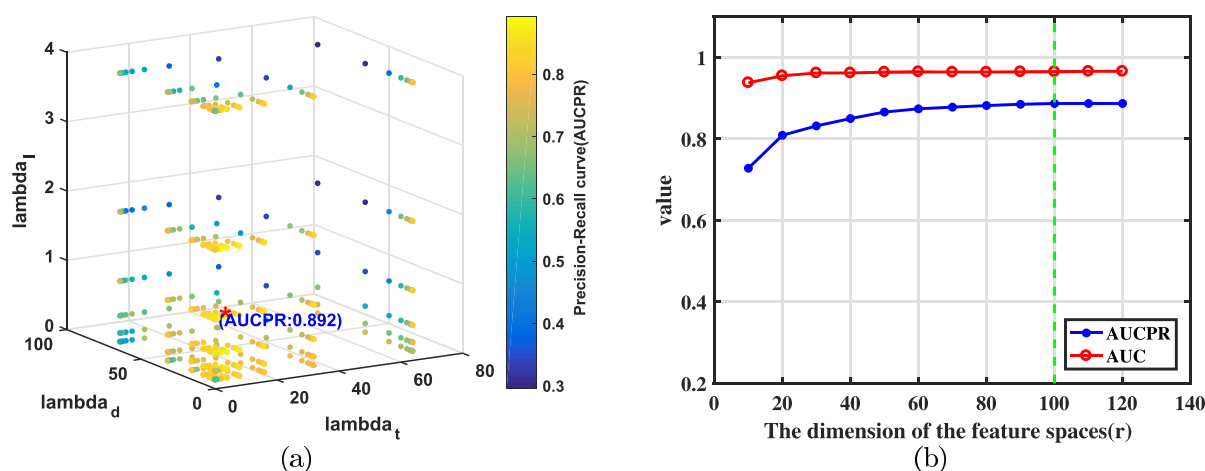


Figure 9. Influences of parameters on the SPLCMF model on the E data set under CVS1. (a) AUCPRs are obtained by different combinations of λ_l , λ_d , and λ_t . The optimal hyperparameters are $\lambda_l = 1$, $\lambda_d = 0.0625$, and $\lambda_t = 1$. (b) AUCs and AUCPRs are obtained by assigning different r values.

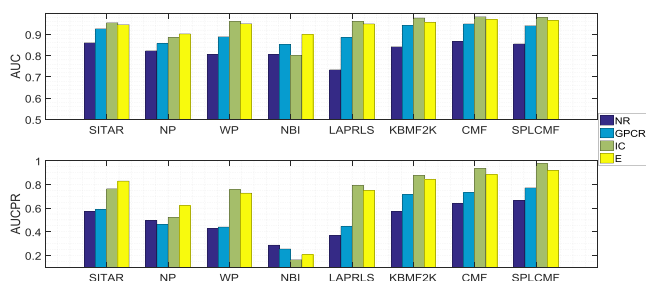


Figure 10. AUCs and AUCPRs averaged over 50 runs by different methods under CVS1.

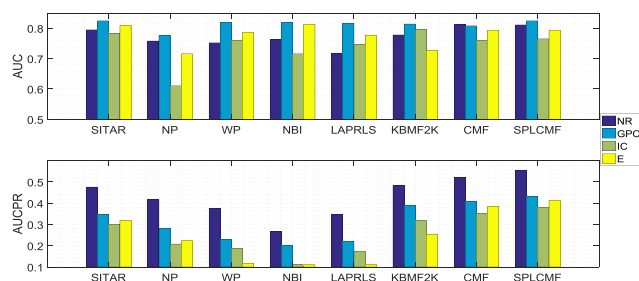


Figure 11. AUCs and AUCPRs averaged over 50 runs by different methods under CVS2.

CVS3: In Figure 5c, a new target may be interacted with one or several drugs. During experiments, in each round, 90% of columns in Y are regarded as training data and the remaining 10% of columns as test data.

Parameter Settings. As mentioned above, our proposed SPLCMF method has four parameters: the dimension of feature space r and the regulation parameters λ_l , λ_d , and λ_t . The setting of parameters of our proposed methods is as follows. The dimension of the feature space r is selected from $\{10, 20, 30,$

$\dots, 120\}$. In our proposed SPLCMF method, we set the regularization parameter λ_l from $\{2^{-3}, 2^{-2}, \dots, 2^2\}$ and $\lambda_d = \lambda_t$ from $\{2^{-4}, 2^{-3}, \dots, 2^6\}$. Thus, we need to choose a better parameter on a different data set. As far as we know, grid search and manual search are widely used in optimization strategies.⁴⁶ In this work, we apply grid search to search for a better parameter.

In order to illustrate the process of grid search simply, we consider only drug–target pair prediction under CVS1 on

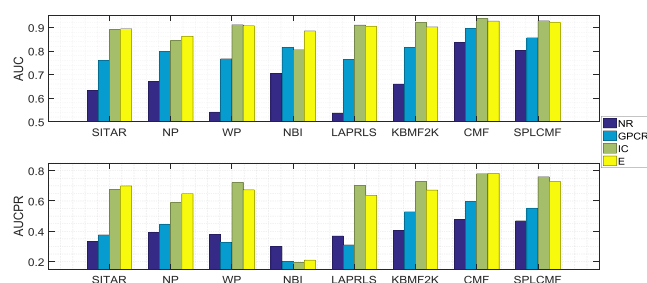


Figure 12. AUCs and AUCPRs averaged over 50 runs by different methods under CVS3.

different data sets. Besides, the dimension of the feature space r is independent of regulation parameters λ_i , λ_d , and λ_t . First, we set the dimension of the feature to $r = 100$ with different combinations of λ_i , λ_d , and λ_t .⁴⁴ The report is shown in Figures

6a, 7a, 8a, and 9a. The red stars are the optimal hyperparameters for SPLCMF with the optimal solution in terms of AUCPRs. Then, we fix the optimal hyperparameters and test the influence of the dimension of feature space r . The results are exhibited in Figures 6b, 7b, 8b, and 9b. In terms of AUCPRs and AUCs, for the convenience of doing the following experiments, we pick the dimension of the feature spaces $r = 100$ (the green line) among all data sets. On the basis of above discussion, we can adopt these parameters for SPLCMF in the following experiments. Of note, the strategies of estimating parameters of drug prediction under CVS2 and target prediction under CVS3 are the same as those of drug–target prediction under CVS1.

Performance Results. In this section, we evaluate the performance of the proposed SPLCMF method. The competing methods are designed for DTI prediction: SITAR, NP, WP, NBI, LAPRLS, KBMF2K, and CMF. Figures 10–12 show that the results are obtained by 5× 10-fold CV for different methods

Table 3. Top Ten Predicted Interactions by SPLCMF on NR, GPCR, IC, and E Data Sets under CVS1^a

	rank	pair	annotation	score	databases		rank	pair	annotation	score	databases
NR	1	D01132	Tazarotene	0.738	K,D,C	GPCR	1	D02149	Epinephrine bitartrate	0.847	K,D
		hsa5915	RARB					hsa151	ADRA2B		
	2	D00316	Etretinate	0.736	K,D,C		2	D02149	Epinephrine bitartrate	0.778	K,D
		hsa5916	RARG					hsa146	ADRA1D		
	3	D00094	Tretinoin	0.733	K,C		3	D00454	Olanzapine	0.777	D,C
		hsa5914	RARA					hsa147	ADRA1B		
	4	D00094	Tretinoin	0.66	K,D,C		4	D00609	Prazosin hydrochloride	0.775	D
		hsa5916	RARG					hsa151	ADRA2B		
	5	D00094	Tretinoin	0.657	K,C		5	D00996	Epinephrine hydrochloride	0.770	K,D
		hsa5915	RARB					hsa148	ADRA1A		
IC	6	D00187	Ergocalciferol	0.647	K,D,C	E	6	D01603	Bethanidine sulfate	0.760	
		hsa7421	VDR					hsa146	Adrenoceptor alpha 1D		
	7	D00316	Etretinate	0.632	K,C		7	D00513	Pindolol	0.740	D,C
		hsa6257	RXRB					hsa147	ADRA1B		
	8	D00316	Etretinate	0.607	K,C		8	D02237	Guanethidine sulfate	0.738	
		hsa6256	RXRA					hsa146	ADRA1D		
	9	D00316	Etretinate	0.600	K,D,C		9	D01603	Bethanidine sulfate	0.736	
		hsa5915	RARB					hsa147	Adrenoceptor alpha 1B		
	10	D01132	Tazarotene	0.598	K,D,C		10	D00451	Sumatriptan	0.734	D,C
		hsa5916	RARG					hsa3351	HTR1B		
E	1	D02356	Verapamil	1.085			1	D01276	Atazanavir sulfate	0.827	
		hsa6329	SCN4A					hsa7363	UGT2B4		
	2	D02356	Verapamil	1.032			2	D01276	Atazanavir sulfate	0.825	
		hsa59284	CACNG7					hsa7365	UGT2B10		
	3	D00332	Gabapentin	1.018			3	D01276	Atazanavir sulfate	0.822	
		hsa5310	PKD1					hsa10720	UGT2B11		
	4	D00629	Nitrendipine	1.005	K,D,C		4	D01276	Atazanavir sulfate	0.820	
		hsa776	CACNA1D					hsa7367	UGT2B17		
	5	D00512	Phenytoin	0.998			5	D01276	Atazanavir sulfate	0.820	
		hsa784	CACNB3					hsa7364	UGT2B7		
K	6	D00332	Gabapentin	0.988			6	D01276	Atazanavir sulfate	0.815	
		hsa3710	ITPR3					hsa54600	UGT1A9		
	7	D00618	Nisoldipine	0.985	K,D		7	D01276	Atazanavir sulfate	0.814	
		hsa776	CACNA1D					hsa54576	UGT1A8		
	8	D00437	Nifedipine	0.985	K		8	D01276	Atazanavir sulfate	0.812	
		hsa778	CACNA1F					hsa7366	UGT2B15		
	9	D01295	Cinnarizine	0.967			9	D01276	Atazanavir sulfate	0.808	
		hsa3708	ITPR1					hsa54577	UGT1A7		
	10	D00560	Pimozide	0.965			10	D01276	Atazanavir sulfate	0.805	
		hsa55503	TRPV6					hsa54575	UGT1A10		

^aInteractions reported in ChEMBL, DrugBank, and KEGG are marked with C, D, and K, respectively. The confirmed DTI pairs are in bold.

Table 4. Fractions of True DTIs among the Predicted Top N (N = 10, 30, 50) Interactions Averaged over 50 Runs^a

		top 10	top 30	top 50		top 10	top 30	top 50
NR	SITAR	4%(5.5%)	3.3%(0%)	2.4%(0.9%)	GPCR	10%(0%)	4.7%(1.9%)	3.6%(0.9%)
	NP	66%(5.5%)	58%(1.8%)	59.2%(2.3%)		100%(0%)	83.3%(2.4%)	79.2%(2.3%)
	WP	68%(4.5%)	54.7%(3%)	51.2%(1.8%)		92%(4.5%)	84.7%(1.9%)	86.4%(3.8%)
	NBI	32%(8.4%)	28.7%(8.4%)	26%(2%)		52%(4.5%)	60%(5.3%)	52.4%(3%)
	LAPRLS	16%(11.4%)	10.6%(7.2%)	9.6%(7%)		92%(4.5%)	86.7%(0%)	86.8%(1.8%)
	KBMF2K	8%(4.5%)	4.6%(3%)	5.2%(1.8%)		0%(0%)	0%(0%)	0%(0%)
	CMF	88%(4.5%)	84%(2.8%)	79.6%(3.6%)		96%(5.5%)	92%(3%)	90.8%(2.3%)
IC	SPLCMF	100%(0%)	86%(2.8%)	78.4%(1.7%)	E	100%(0%)	100%(0%)	98.4%(2.2%)
	SITAR	0%(0%)	0%(0%)	0%(0%)		0%(0%)	0%(0%)	0%(0%)
	NP	100%(0%)	94%(1.5%)	93.6%(2.6%)		100%(0%)	92.7%(2.8%)	93.2%(1.8%)
	WP	100%(0%)	98.7%(1.8%)	98%(2%)		92%(4.5%)	94%(2.8%)	96%(1.4%)
	NBI	48%(8.4%)	50.7%(2.8%)	57.2%(1.8%)		90%(0%)	96.7%(0%)	89.6%(2.2%)
	LAPRLS	100%(0%)	100%(0%)	98%(1.4%)		90%(7.1%)	93.3%(2.4%)	95.2%(1.8%)
	KBMF2K	0%(0%)	0%(0%)	0%(0%)		0%(0%)	0%(0%)	0%(0%)
	CMF	100%(0%)	100%(0%)	100%(0%)		100%(0%)	98%(1.8%)	96.4%(0.9%)
	SPLCMF	100%(0%)	100%(0%)	100%(0%)		100%(0%)	98.7%(1.8%)	98.4%(0.9%)

^aThe best results are highlighted in bold under CVS1. SDs are given in parentheses.

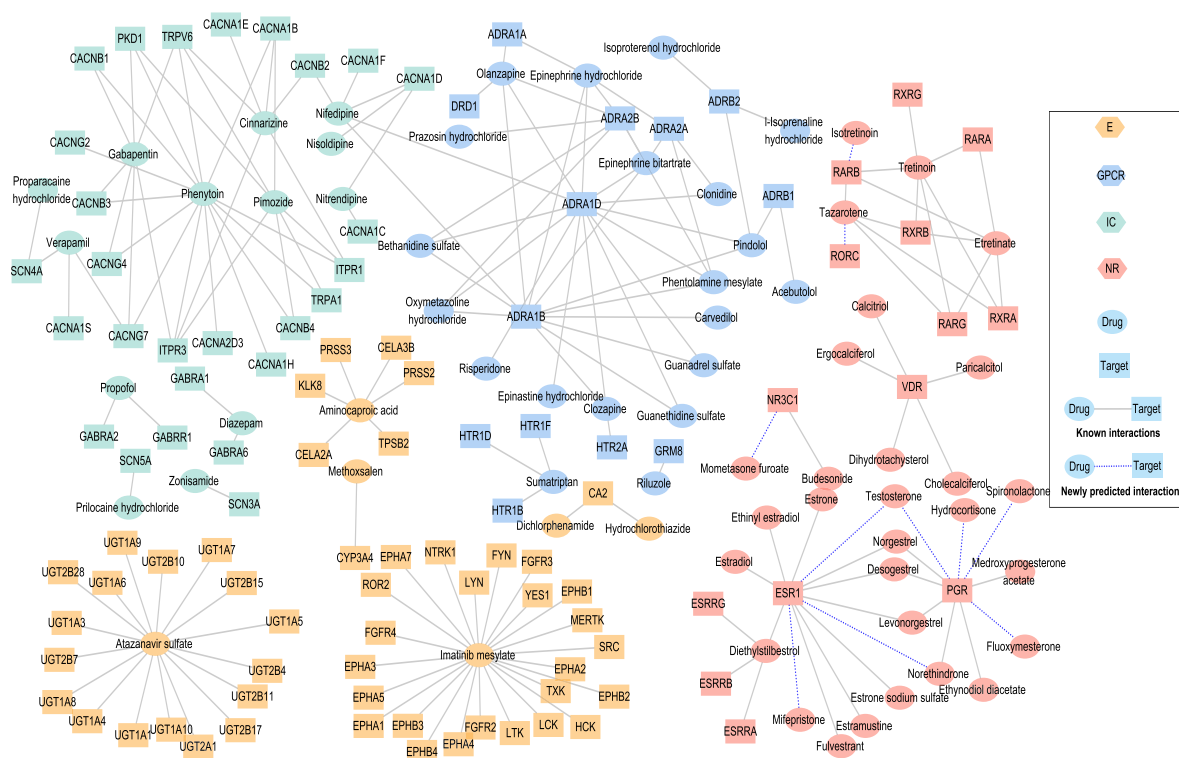


Figure 13. Predicted NR, GPCR, IC, and E interaction networks. Hexagons, circles, and rectangles represent data sets, drugs, and target proteins, respectively. Gray and blue lines indicate known interactions and newly predicted interactions with the top 50 highest scores, respectively.

under CVS1, CVS2, and CVS3. For drug–target pair prediction under CVS1, CMF achieves the best AUC values among all methods. However, SPLCMF outperforms all seven competing methods for four data sets (NR, GPCR, IC, and E) in terms of AUCPRs. For example, for the GPCR data set, the AUCPRs of SPLCMF, CMF, KBMF2K, SITAR, NP, LAPRLS, WP, and NBI rank in first, second, third, fourth, fifth, sixth, seventh, and eighth places. Moving to drug prediction under CVS2, the AUC values of SPLCMF, CMF, KBMF2K, and NBI are the highest in GPCR, NR, IC, and E data sets, respectively. On the other hand, as for AUCPRs, similar to the CVS1, SPLCMF outperforms other methods in all cases. Finally, we reach the results of target prediction under CVS3. CMF achieves the best results among all

methods in all four data sets. That is, our proposed SPLCMF method is not suitable for target prediction, while incorporating self-paced learning into CMF might not be necessarily useful for improving the performance in this case. To conclude, our proposed SPLCMF method is a competitive approach in pair prediction and drug prediction, except target prediction.

Table 3 shows the top 10 interactions predicted by our proposed SPLCMF on the NR, GPCR, IC, and E data sets. We give the top 10 pairs with respect to scores in the fourth column of the table for each data set. Then, the names of the drugs and targets are listed in the third column. In addition, we apply ChEMBL, DrugBank, and KEGG to demonstrate the effectiveness of our method. These databases are marked C,

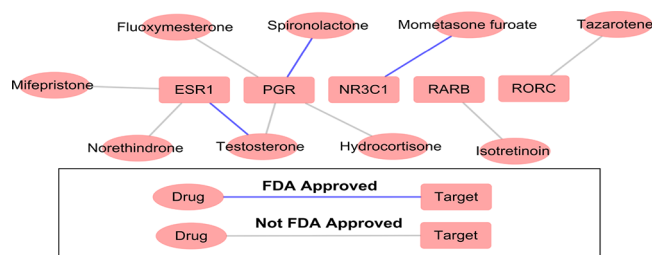


Figure 14. Newly predicted interactions are extracted from NR, GPCR, IC, and E interaction networks. The circles and rectangles represent drugs and target proteins, respectively. Gray and blue lines indicate not-FDA-approved drug target genes and FDA-approved drug target genes, respectively.

Table 5. Results Averaged over 50 Runs^a

scenario	methods	AUC	AUCPR
CVS1	CMF	0.931(0.001)	0.639(0.003)
	SPLCMF	0.933(0.002)	0.733(0.003)
CVS2	CMF	0.828(0.004)	0.376(0.004)
	SPLCMF	0.831(0.005)	0.491(0.006)
CVS3	CMF	0.884(0.004)	0.529(0.012)
	SPLCMF	0.886(0.003)	0.456(0.007)

^aThe best results are highlighted in bold. SDs are given in parentheses.

D, and K. For example, for the NR data set, the top 10 pairs are reported in at least one of these databases. Our proposed SPLCMF identifies 3 and 7 out of the top 10 interactions predicted on IC and GPCR data sets. Of note, the predicted interactions may also exist in reality but have not yet been reported.

Table 4 exhibits the fractions of true DTIs among the top $N(N = 10, 30, 50)$ predictions implemented by each DTI method under CVS1. Compared with competing methods, our proposed SPLCMF achieves better results for all cases except the top 50 interactions in the NR data set. For example, the top 30 interactions predicted by our proposed SPLCMF are 86, 100, 100, and 98.7% for NR, GPCR, IC, and E datasets, respectively. In addition, Figure 13 indicates known interactions and newly predicted interactions with the 50 highest scores by our proposed SPLCMF for NR, GPCR, IC, and E data sets in the Supporting Information. To explore the DTI pairs that are selected by our proposed SPLCMF method in depth, the newly predicted interactions (blue line) are extracted from Figure 13. As shown in Figure 14, the network is constructed by Cytoscape by integrating the biological interaction from DrugBank, including 9 drugs and 5 targets. We can find that ESR1 and PGR are interacted with several drugs. Besides, three drug target genes are approved by the FDA. The blue line indicates FDA-approved drug target genes. On the basis of discussions, our proposed SPLCMF method can help biologists significantly reduce the cost of biological experiments.

In addition to using NR, GPCR, IC, and E data sets for testing, we also validate our proposed SPLCMF with an additional benchmark data set compiled by Kuang et al.,³⁶ which is a larger data set with 3681 known interactions including 786 drugs and 809 targets than the above-mentioned four benchmark data sets. On the basis of the discussion for NR, GPCR, IC, and E data sets, we find that SPLCMF and CMF are superior to other methods. Therefore, in an additional benchmark data set, we compare only the performance of CMF and SPLCMF. The

comparison results of the two methods are shown in Table 5; although our proposed SPLCMF outperforms CMF by around 1% in terms of AUCs in all scenarios, the gaps in AUCPR between SPLCMF and CMF can be clearly seen. The proposed SPLCMF method is attained significantly by around 11% compared with CMF in CVS1 and CVS2. Therefore, our SPLCMF method is a competitive method for pair prediction and drug prediction in addition to target prediction.

DISCUSSION AND CONCLUSIONS

Due to the presence of noise and missing data, most of the existing MF methods can easily fall into bad local minima, where MF models are generally nonconvex. This work presents a novel MF framework, namely, SPLCMF for identifying unknown DTI pairs. The novelty of SPLCMF can be attributed to a combination of several factors. First, CMF projects drugs and targets into a common low-rank feature space and then integrates the drug chemical structure information, protein sequence information, and known DTI information into regularized least-squares. Moreover, SPL includes DTI matrix elements in CMF training from easy to complex to further improve the prediction ability of the model and can be effective in avoiding bad local minima, especially in the presence of heavy noises and missing data. Furthermore, soft weighting is a method of assigning real weights, which more faithfully reflects the potential importance of samples in training. In a word, the framework of SPLCMF can provide a useful tool to predict unknown DTIs, which may provide new insights into drug discovery, drug side-effect prediction, and repositioning existing drugs.

To evaluate the effectiveness of the proposed SPLCMF method, an extensive set of experiments are performed on synthetic data and five benchmark data sets, compared with seven state-of-the-art DTI prediction methods. Through comprehensive tests, these results indicate that SPLCMF can provide a useful tool to predict new DTIs and repurpose existing drugs. For example, on the basis of synthetic data, SPLCMF is robust to noise and missing data. Then, on average, SPLCMF achieves the best under CVS1 and CVS2 in terms of common machine learning evaluation metrics, except the AUC value. Under CVS3, SPLCMF attains the second best among all evaluation metrics. The reason is lack of target–target similarity network information. These results show that our proposed SPLCMF outperforms other existing state-of-the-art approaches in predicting new pairs and new drugs and is comparable or even better than existing methods in predicting new targets. In future work, we plan to integrate more than one similarity matrix over drugs as well as those over targets for further improvement on the performance of the drug–target prediction methods²⁶ and collect more test data sets to suppose our proposed method.⁴⁷

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jcim.9b00408.

Top 50 predicted interactions of SPLCMF-CVS1-NR, SPLCMF-CVS1-GPCR, SPLCMF-CVS1-IC, and SPLCMF-CVS1-E (PDF)

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Author Contributions

L.Y.X. and Z.Y.Y. designed the simulations. L.Y.X. and H.Z. provided the mathematical proof. L.Y.X., Z.Y.Y., and Y.L. contributed to collect data sets and analyze data. L.Y.X. and Y.L. designed and implemented the algorithm. L.Y.X., Z.Y.Y., H.Z., and Y.L. contributed to the interpretation of the results. L.Y.X. took the lead in writing the manuscript. L.Y.X., Z.Y.Y., H.Z., and Y.L. revised the manuscript.

Notes

The authors declare no competing financial interest.

The all data and source code are available from <https://github.com/Macau-LYXia>.

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ABBREVIATIONS USED

DTIs	drug–target interactions
SPL	self-paced learning
CMF	collaborative matrix factorization
E	enzymes
IC	ion channels
GPCR	G protein-coupled receptors
NR	nuclear receptors
NP	nearest profile
WP	weighted profile
BLM	bipartite local model
SVM	support vector machine
LapRLS	Laplacian regularized least-squares
NBI	network-based inference
SITAR	similarity-based inference of drug–targets
MF	matrix factorization
KBMF2K	kernelized Bayesian matrix factorization with twin kernels
MSCMF	multiple similarities collaborative matrix factorization
NRLMF	Neighborhood Regularized Logistic Matrix Factorization
GRMF	graph regularized matrix factorization
WGRMF	weighted graph regularized matrix factorization

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